13

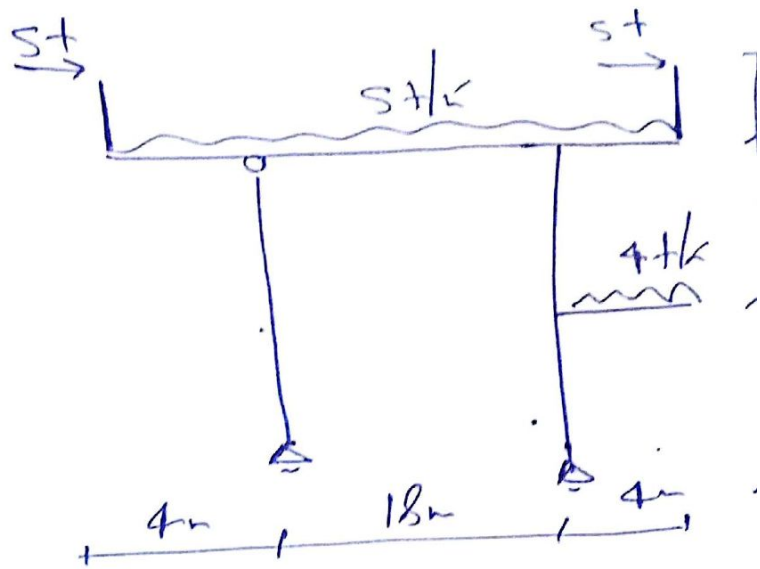
frames

$f_c = 250\ kg/cm^2$

$f_y = 3600\ kg/cm^2$

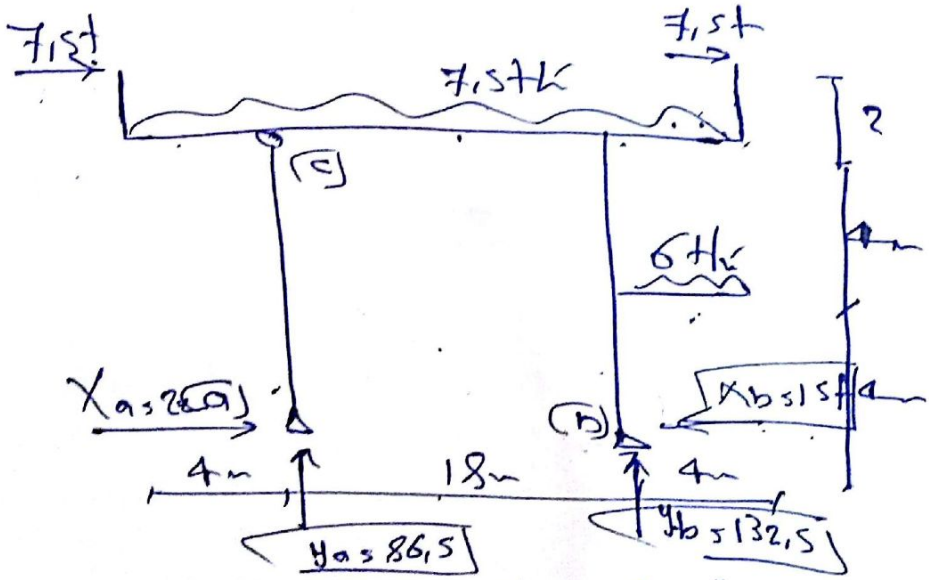
breadth of frame = 35

all load is working load



solution

ultimate load ، 1.5 ، ثم نضرب الأحمال في



$\Sigma M_{on\ A} = 0$

$7,5 \times 18 \times 9 + 7,5 \times 4 \times (2+18) - 7,5 \times 4 \times 2$

$+ 7,5 \times 10 + 7,5 \times 6 + 6 \times 4 (2+18) - 18 y_b = 0$

$y_b = 132,5\ ton$

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$\Sigma (M_b)_+ = 0$ [2]

$7{,}5 \times 4 \times 2 - 7{,}5 \times (18+4) \times 11 + 7{,}5 \times 10 + 7{,}5 \times 1 \ldots$

$+ 6 \times 4 \times 2 + 18 y_a = 2 \ldots$

$y_a = 86{,}5$ ton

$\uparrow + \Sigma f_y = 2 \ldots$

$y_a + y_b - 7{,}5(4+18+4)$

$86{,}5 \quad 132{,}5$

$- 6 \times 4 = 2 \ldots$ OK

$\Sigma f_x = 0$

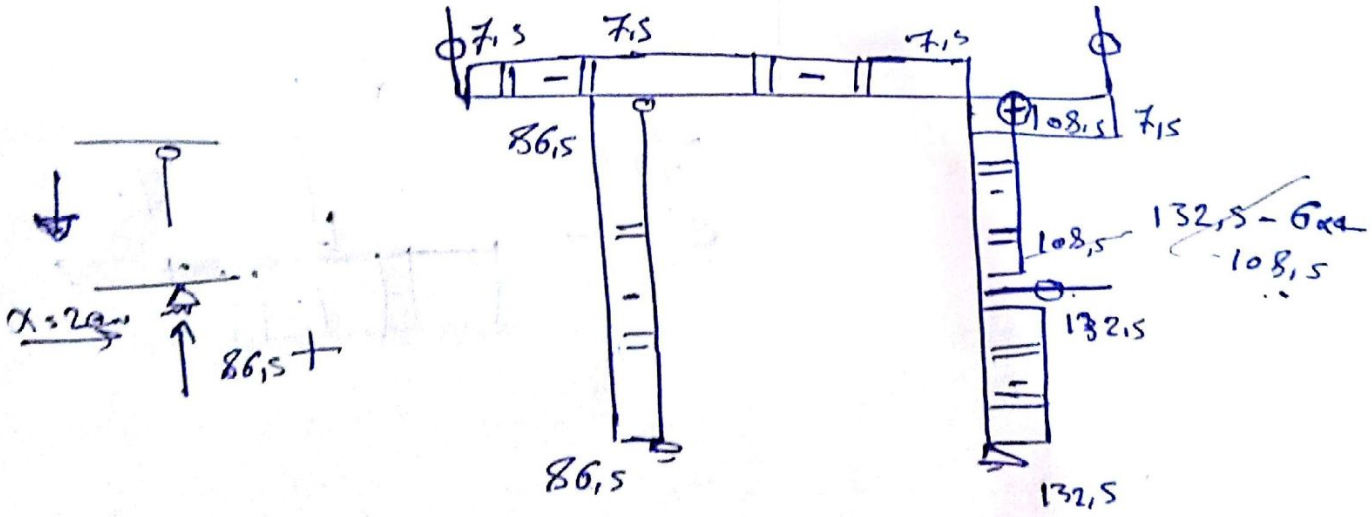
$\rightarrow +$

$7{,}5 + 7{,}5 - X_b = 2 \ldots$

$X_b = 15$ ton

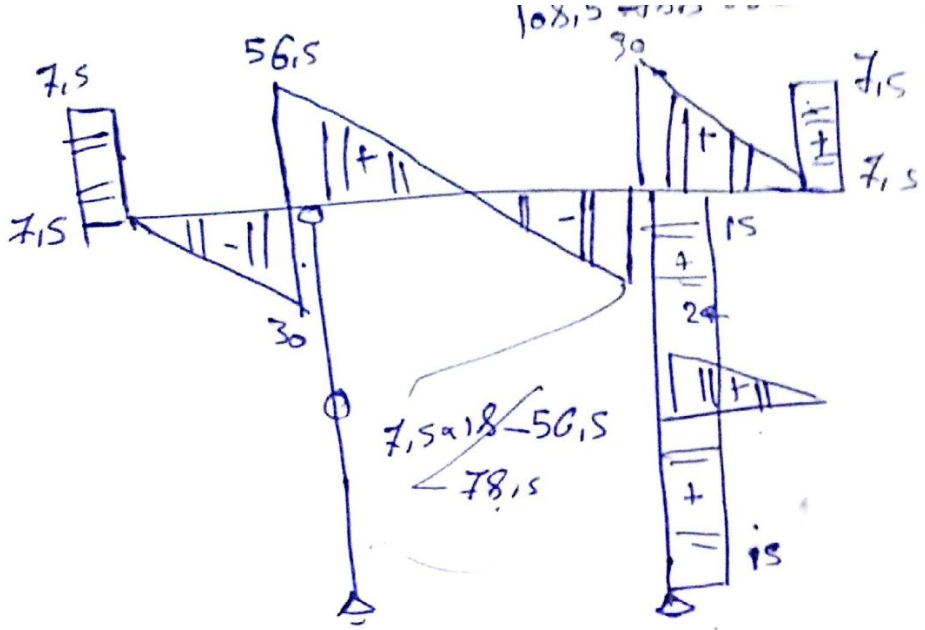
———————————

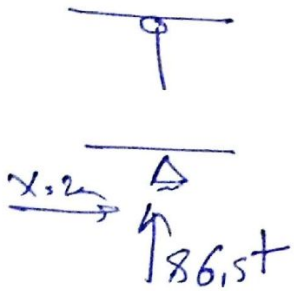
for N.F.D



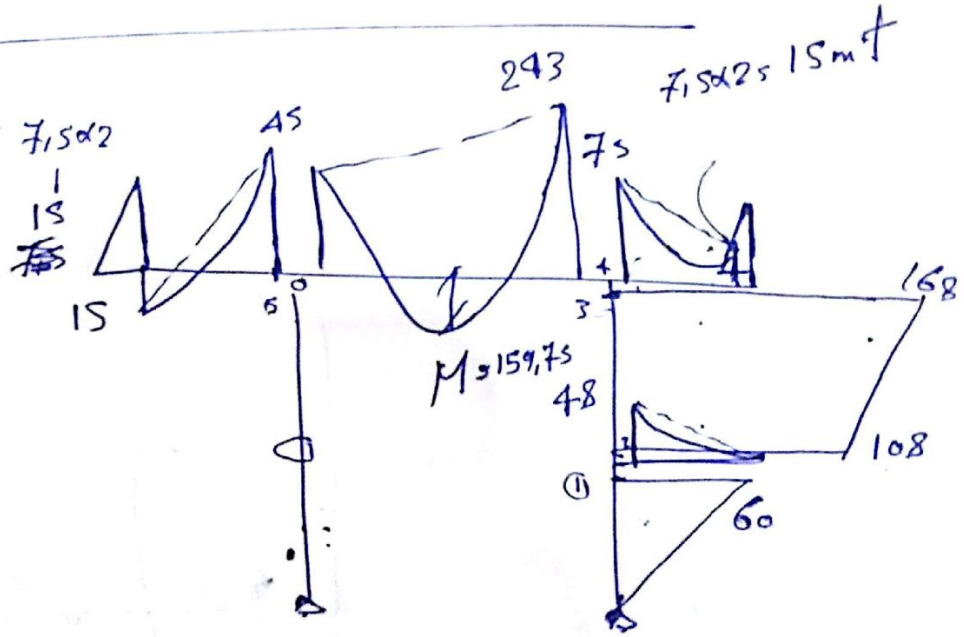
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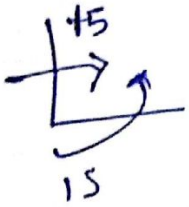
S.F.D





for B.M.D



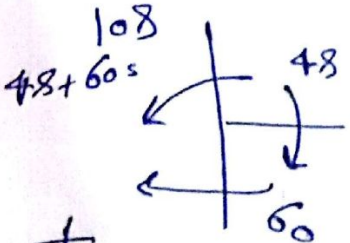


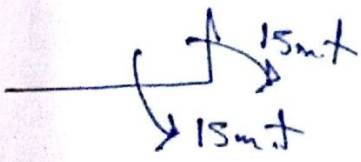
$M_1 = 15 \times 4 = 60$ t.m

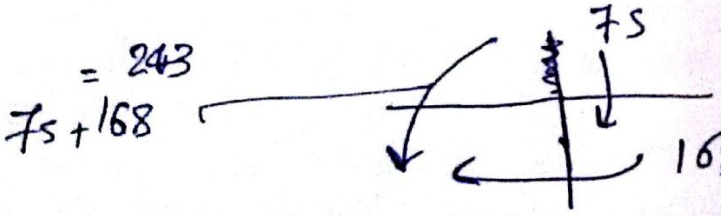
$M_2 = 6 \times 4 \times 2 = 48$ m.t

$M_3 = 15 \times 8 + 6 \times 4 \times 2 = 168$ t.m

$M_4 = \underset{\text{uniform}}{7,5 \times 4 \times 2} + \underset{\text{Concentrated}}{7,5 \times 2} = 75$ t.m

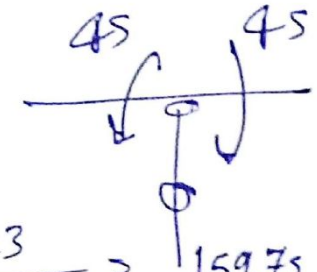






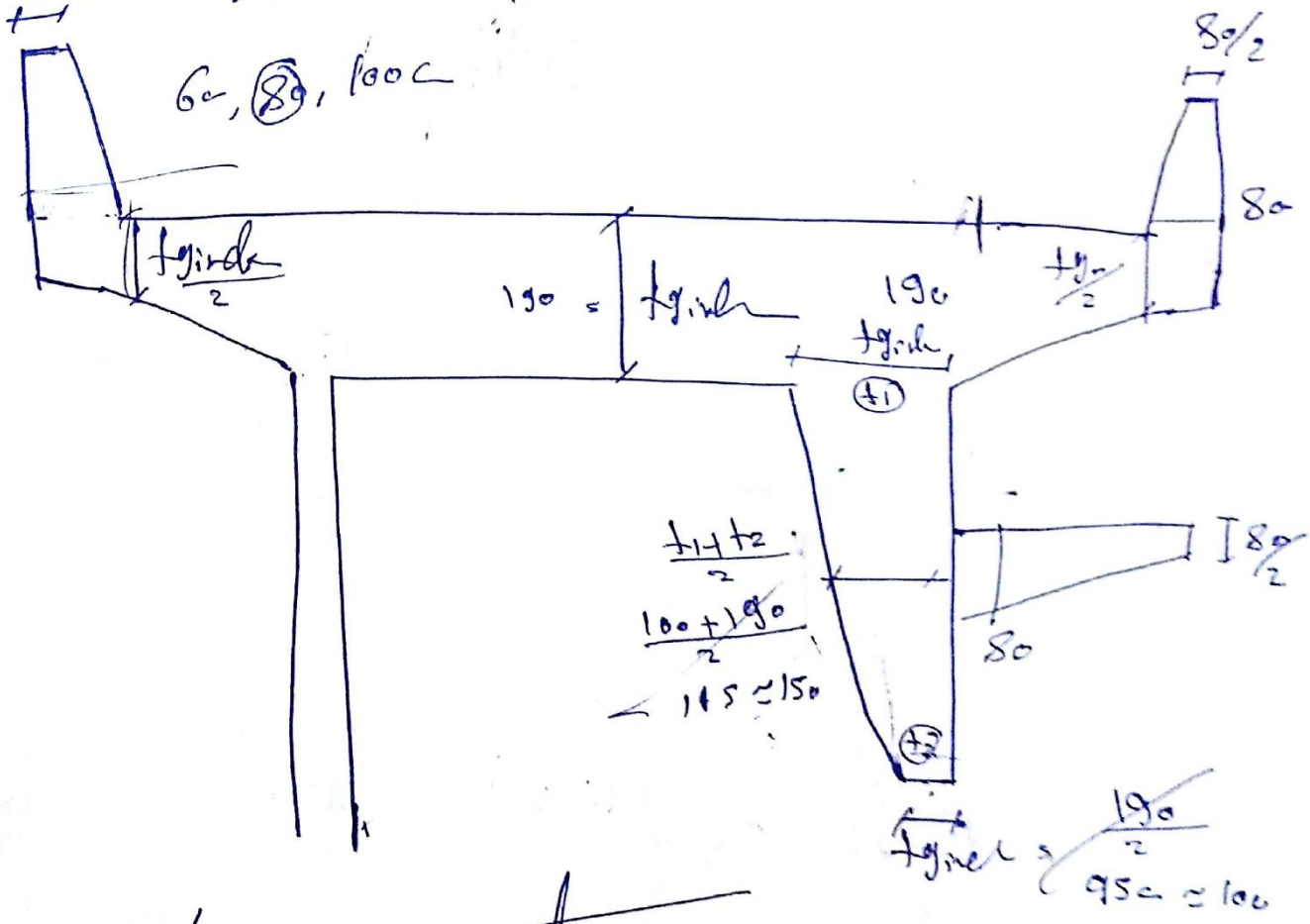
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$M_S = 7{,}5 \times 4 \times 2 - 7{,}5 \times 2{,}5 = 45\ mt$



$$\#M = \frac{w l^2}{8} - \frac{45 + 243}{2}$$

$$M = \frac{7{,}5 \times 18^2}{8} - \frac{45+243}{2} = 159{,}75\ mt$$



$M_{max} = 243\ mt$

$$d_0 = c_1 * \sqrt{\frac{M_u d}{f_c a b}}$$

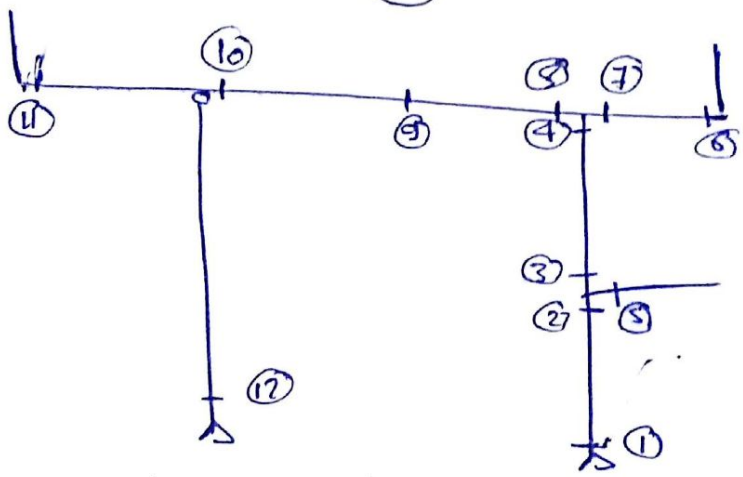
$$d_0 = 3a \sqrt{\frac{243 \times 10^5}{250 \times 35}} = 158{,}1\ cm$$

$$d = 1{,}15\, d_0 = 1{,}15 * 158{,}1 = 181{,}81\ cm$$

$t = 190\ cm$ $\quad d_1 = 185\ cm$

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(5)



sec	M	N	b	t	As Cm^2	choix	Q	stirrups
1	Zero	-1375	35	100	19,8	5#25	15	5Φ8L
2	60	-1325		150	19,8	5#25	15	5Φ8L
3	108	-1085		150	19,8	5#25	15	5Φ8L
4	168	-1085		190	19,8	5#25	15	5Φ8L
5	48	Zero		80	21,5	5#25	15	5Φ8L
6	15	+715		100	19,8	5#25	21	5Φ8L
7	75	+715		190	19,9	5#25	Zero	5Φ8L
8	243	-715		190	44,2	10#25	30	5Φ8L
9	159,75	-715		190	29,04	6#25	78,5	8Φ8L
10	45	-715		190	19,8	5#25	Zero	5Φ8L
11	15	-715		100	19,8	5#25	56,5	6Φ8L
12	Zero	-865	b=35	35	12,75	4#12	Zero	5Φ8L

for Linkmeber sec (12)

$$P_{ult} = 0{,}35 * f_c A_c + 0{,}67 * f_y A_s$$

$$A_s = \frac{1}{100} A_c$$

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$\frac{865 \times 10^3}{kg}$ ton $= 0,35 \times 250 \times Ac + 0,67 \times 3600 \times \frac{1}{100}$

$Ac = 778,5\ cm^2$

$t = \frac{Ac}{b} = \frac{778,5}{35} = 23,2c$

$t < b$

use sec 35×35 c

$As = \frac{1}{100} Ac = \frac{1}{100} \times 35 \times 35 = 12,25 c^2$

4 #T22

sec ①, ②, ③, ④

$P_{Cam\ min} = \frac{0,04 \times fa \times Ac}{1000}$ bad

$= \frac{0,04 \times 250 \times 35 \times 185}{1000} = 66,5$ ton

sec4

$M = 168$ mt $\quad N = -108,5$ ton

392 P

$P_{balal} = 2144 \times \frac{fc}{6000 + \frac{fy}{\gamma s}} \times \frac{bd}{1000}$

$= 2144 \times \frac{250}{6000 + \frac{3600}{1.15}} \times \frac{35 \times 185}{1000}$

$P_{bal} = 380,1$ ton $> P = 108,5$ t

tension failure

$e = \frac{M}{N} = \frac{168}{108,5} = 1.5m$

$e_s = e + \frac{t}{2} - Cave$

$e_s = 1.5 + \frac{1,9}{2} - 0,05 = 2,45$ m

$M_s = N \times e_s = 2,4 \times 108,5 = 260,4$ mt

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$$d_s = c_1 \sqrt{\frac{M_s}{f_{cu} \times b}}$$

$$185 = c_1 \times \sqrt{\frac{260.4 \times 10^5}{250 \times 35}}$$

$c_1 \longrightarrow \delta = 0.826$

$$\# \quad A_s = \frac{M_s}{f_y \times j \times d} - \frac{N_u}{f_y / \gamma_s}$$

to $\longrightarrow$ kg

$$A_s = \frac{260.4 \times 10^5}{3600 \times 0.826 \times 185} - \frac{108.5 \times 10^3}{\frac{3600}{1.15}}$$

$$A_s = 12.67 \ cm^2$$

$$A_{s\,min} = \frac{11}{f_y} b d = \frac{11}{3600} \times 35 \times 185$$

$$= 19.8 \ cm^2 > A_s$$

use $A_{s\,min} = \#25$

for section (5)

$M_y = 48 \ mt$

$N = 2 \ cm$

$t = 80 \ cm$

$$d_s = c_1 \times \sqrt{\frac{M_u}{f_{cu} \times b}}$$

$$75 = c_1 \times \sqrt{\frac{48 \times 10^5}{250 \times 35}}$$

$c_1 \longrightarrow \delta = 0.826$

$$A_s = \frac{M}{f_y \times j \times d} = \frac{48 \times 10^5}{3600 \times 0.826 \times 75} = 21.5 \ cm^2$$

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$A_{sn} = \frac{11}{fy} b d_s \frac{11}{3600} * 35 * 75 = 8,02 cm^2 < A_s$ ok

use $= \#25$

For section 7

$M = 75 mt$ $\quad N = +7,5 ton$

$\# \quad e = \frac{M}{N} = \frac{75}{7,5} = 10 m$

$\# \quad \frac{e}{t} = \frac{10}{1.9} = 7 > 0,5$ big ecc

$\# \quad e_s = e - \frac{t}{2} + Cr$

$= 10 - \frac{1,9}{2} + 0,05 = 9.1 m$

$\# \quad M_s = N . e_s = 7,5 * 9,1 = 68,25 mt$

$\# \quad d = C_1 \sqrt{\frac{M_s}{f_a \alpha b}}$

$185 = C_1 * \sqrt{\frac{68,25 \alpha 10^5}{250 \alpha 35}}$

$C_1 \longrightarrow \delta = 0,826$

$A_s = \frac{M_s}{fy \alpha j \alpha d} + \frac{N_{ult}}{fy / \delta_s}$

$A_s = \frac{68,25 * 10^5}{3600 * 0,826 * 185} + \frac{7,5 \alpha 10^3}{\frac{3600}{1.15}} = 14.8 cm^2$

$A_{s\,min} = \frac{11}{fy} b d_s \frac{11}{3600} * 35 \alpha 185 = 19,8 cm^2$

$= \#25$

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fonsec 10, sec 11

H = 45 mt

N = -7,5 t

$$d = c_1 \times \sqrt{\frac{M_u t}{f_c \times b}}$$

$$185 = c_1 \times \sqrt{\frac{45 \times 10^5}{250 \times 35}}$$

$c_1 \longrightarrow \delta = 0,826$

$$A_s = \frac{M_u t}{f_y \text{ajd} \times d} = \frac{45 \times 10^5}{3600 \times 0,826 \times 185}$$

$A_s = 8,18 \text{ cm}^2$

$A_{s\min} = 14,8 \text{ cm}^2$ ⟹ 5 #25

---

sec 8

H = 243 mt

N = -7,5 t

$$d = c_1 \times \sqrt{\frac{M_u t}{f_c \times b}}$$

$$185 = c_1 \times \sqrt{\frac{243 \times 10^5}{250 \times 350}}$$

$c_1 \longrightarrow \delta = 0,82$

$$A_s = \frac{M_u t}{f_y \text{ajd}} = \frac{243 \times 10^5}{3600 \times 0,82 \times 185} = 44,2 \text{ cm}^2$$

10 #25

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design sec g

$M_s = 159,75$ kN.m $N_s = 7,5$ t

$$d_s = c_1 * \sqrt{\frac{M_u}{f_y * a_j * d}}$$

$$185 = c_1 * \sqrt{\frac{159,75 * 10^5}{3600 * 0,826 * 185}}$$

$c_1 \longrightarrow \beta = 0,826$

$$A_s = \frac{M_u}{f_y * a_j * d} = \frac{159,75 * 10^5}{3600 * 0,826 * 185}$$

$A_s = 29,04 \text{ cm}^2$

6 #25

check shear ② $\longrightarrow$ $\phi 8$

$q_{str} = q_{ult} - q_a$ cracked

$q_{ult} = q_{str} + q_a$ cracked

$$q_{ult} = \frac{n * A_{str} * f_{yst}}{\gamma_s} + 0,12 \sqrt{\frac{25}{1,5}} * d_0$$

$s = \phi 8$ / $s * a * b$

$$q_{ult} = \frac{2 * 0,503 * \frac{2800}{1,15}}{35 * 20} + 0,12 * \sqrt{\frac{25}{1,5}} * d_0$$

N/mm^2

$= 8,43$ kg/cm^2

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(11)

$$\rho_{ult} = \frac{M_u \times 10^3}{b \times d^2}$$

$$M_u = \frac{\rho_{ult} * b \times d^2}{10^3} = \frac{8,43 * 35 * 185^2}{10^3}$$

$$M_u = 54,58 \text{ t.m}$$

ملحوظات

العتبات التي عليها Compression et tension يتم وضع الحديد على الضيق الخارجي مع مراعاة التماثل ويكون عدد الأسياخ زوجي

A_s الحديد الذي فوق يحمل مقاومة الكبس

و A'_s يحمل (40x — 10x) من A_s الذي فوق بنسبة 30% عادة

3#12 3#16 3#18

سقوط الكمرات إذا كان الانبعاج V.

يتم وضع أسياخ لمقاومة الانكماش

تحسب المسافة الزائدة عن 30 والعدد لا يقل عن #12

$t = 190$ cm

عدد المسافات = $\frac{190}{30} = 6,33$

يقرب لأقرب رقم صحيح بالزيادة

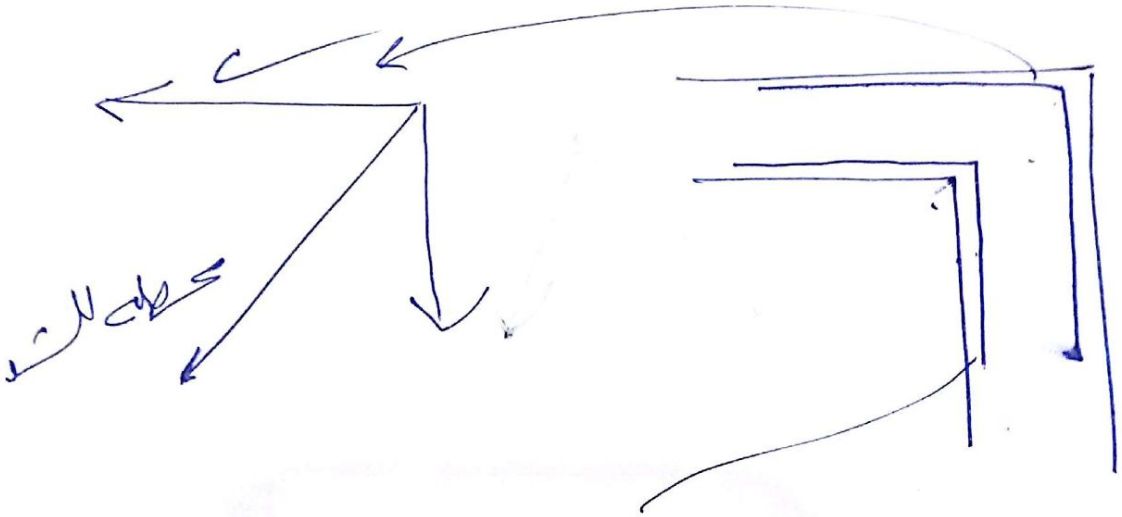
مسافات 7 = عدد المسافات

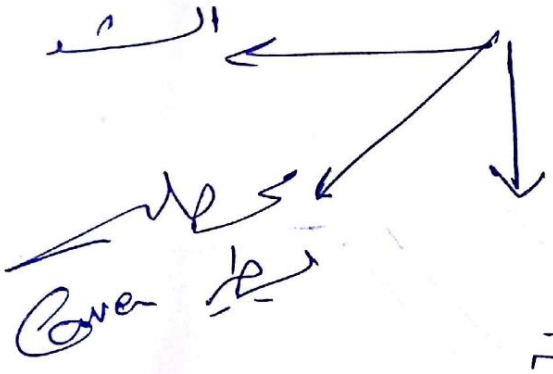
عدد الأسياخ = عدد المسافات + 1 = 7 + 1 = 8

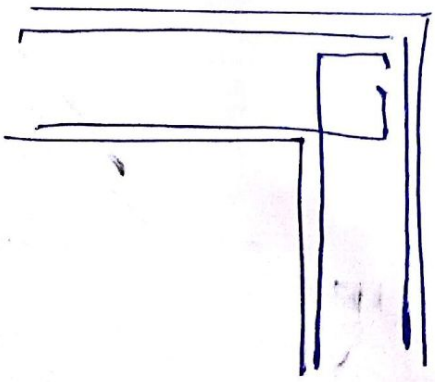
Shi Kayen

ملحوظة عند وضع حديد التسليح السفلي الخاص
بالـ Girder يراعى ان $\frac{1}{3}$ الحديد لا يتم لعدى
من اول السيخة

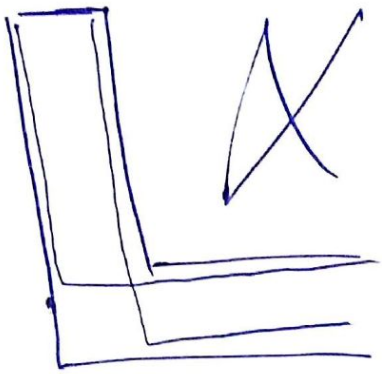
ملحوظة خاصة على الشد

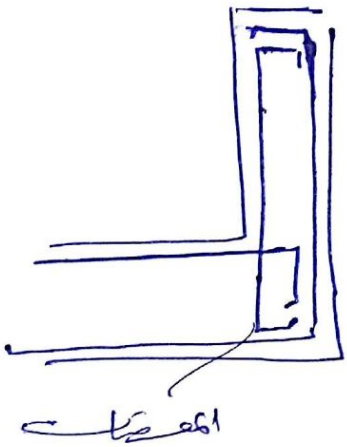


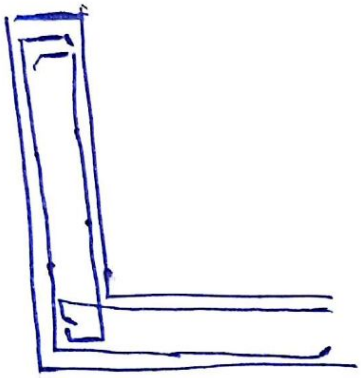


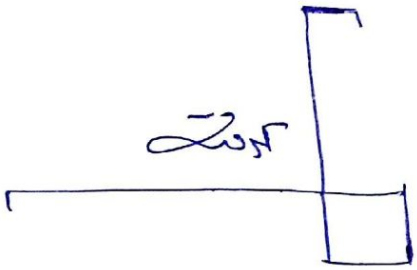


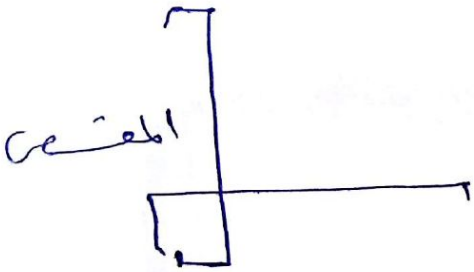
Scanned by CamScanner

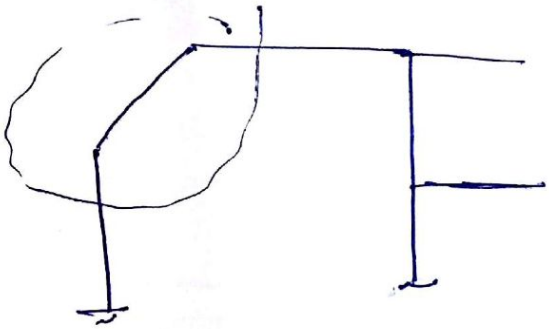


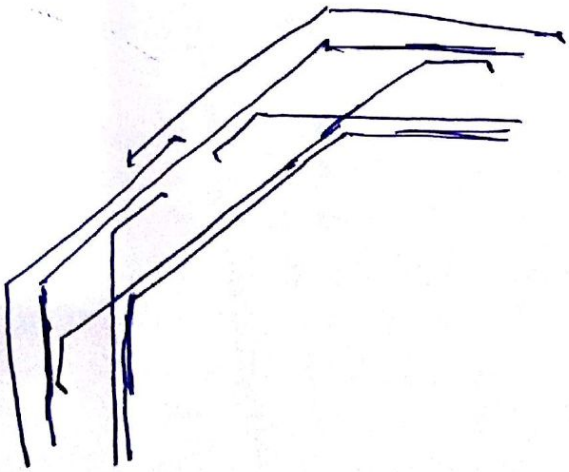




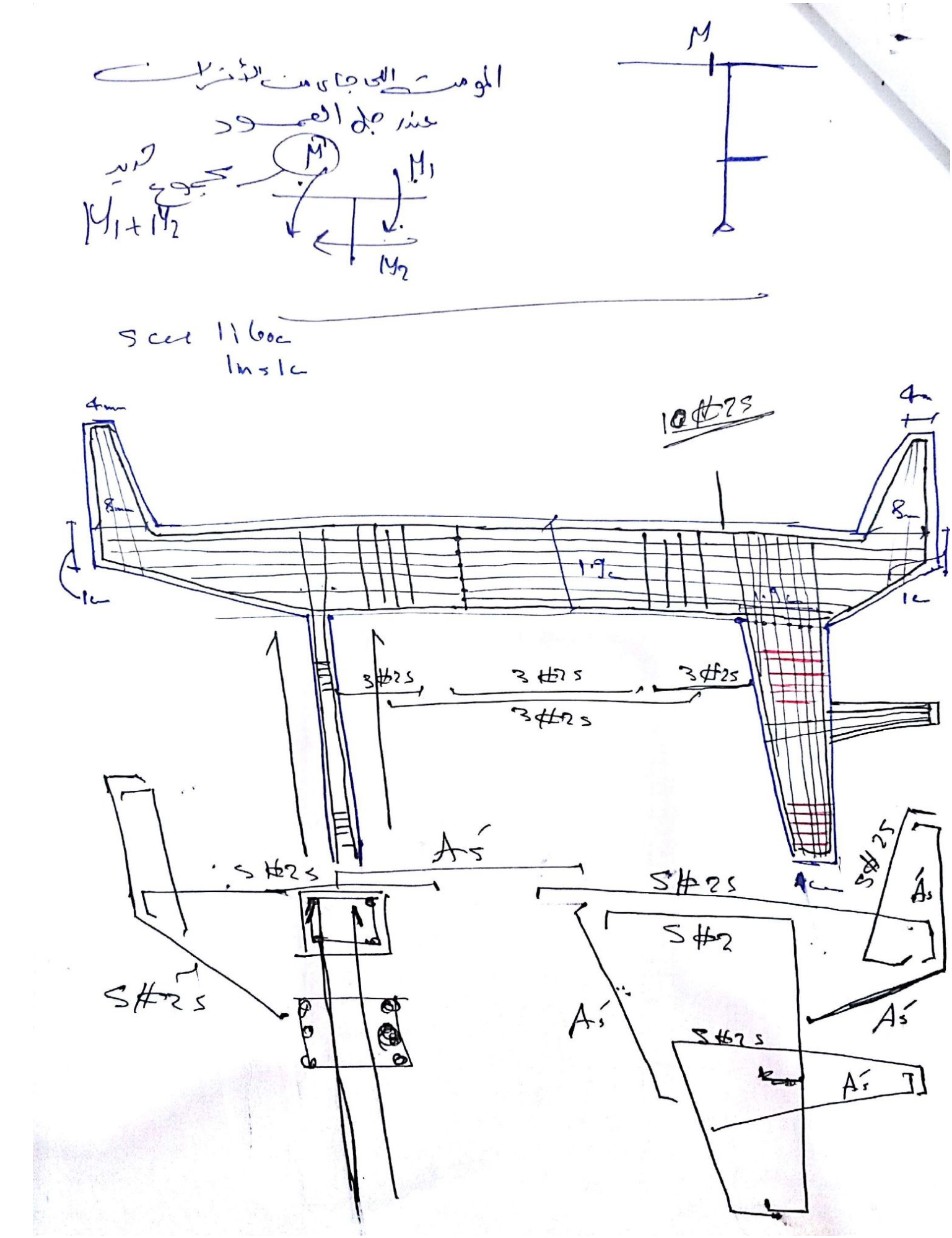




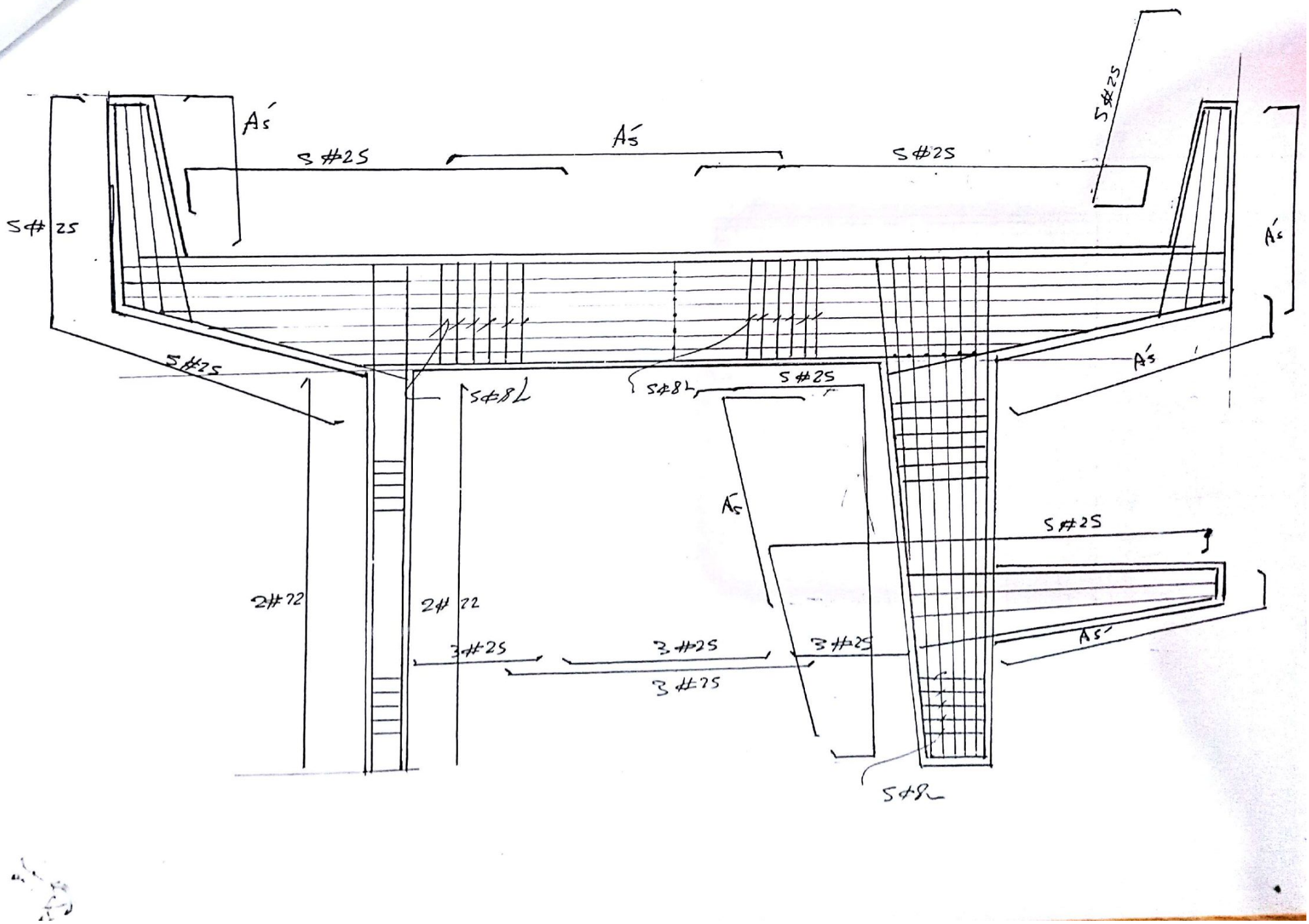




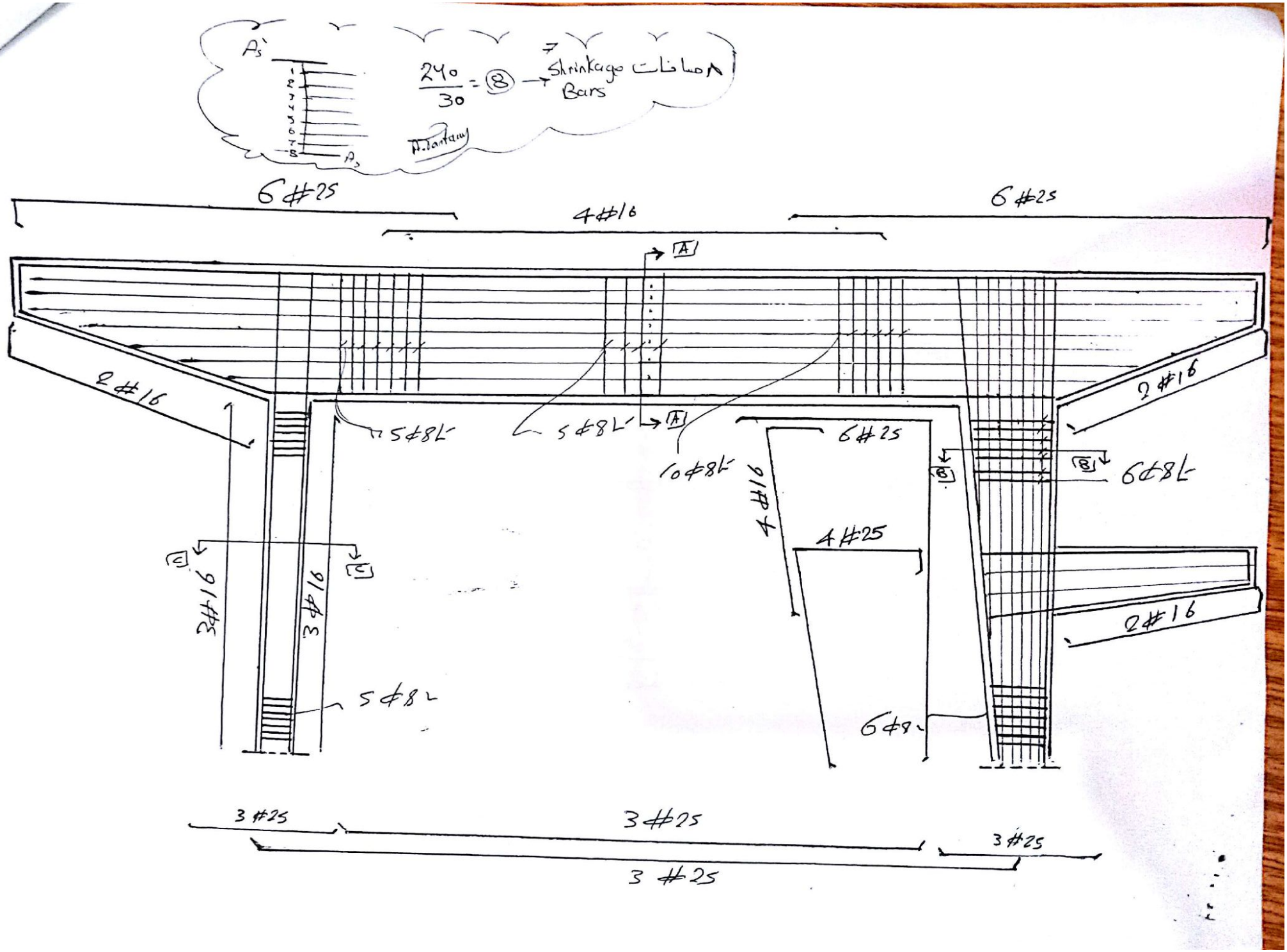
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Sections

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